

A Hybrid RF-Based Off-Grid Communication Network with Opportunistic Cloud Synchronization for Real-Time Traveller Safety and Emergency Response

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Project Overview

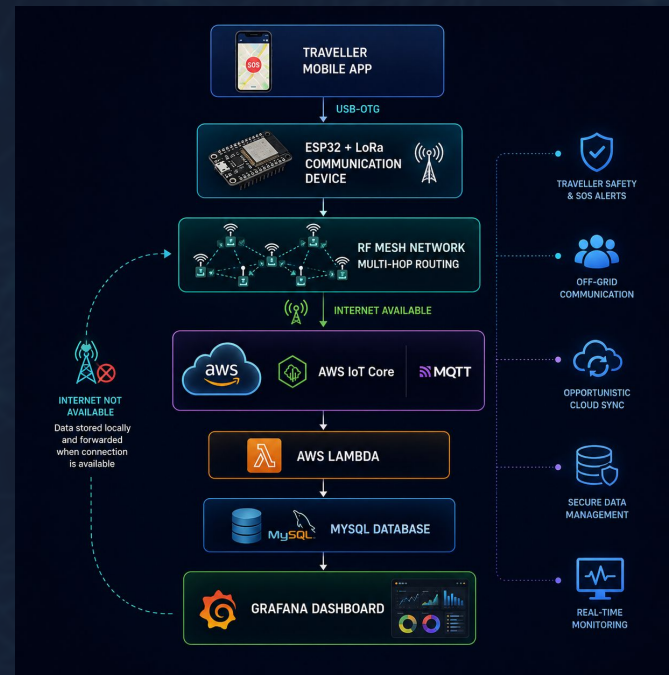
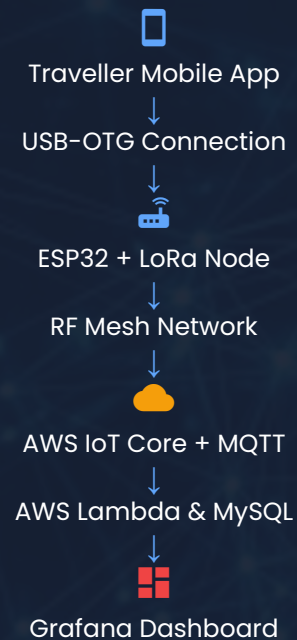
CHALLENGE

- Lack of cellular coverage in remote environments
- Emergency communication failures
- Limited traveller safety monitoring
- Dependence on internet infrastructure

TECHNOLOGY STACK

Flutter	ESP32	LoRa
AWS IoT Core	MQTT	
MySQL	Grafana	

SYSTEM ARCHITECTURE



OBJECTIVES

- Develop RF mesh communication
- Mobile app and IoT integration
- SOS emergency alert propagation
- Opportunistic cloud synchronization
- Real-time monitoring platform

EXPECTED OUTCOMES

- Reliable off-grid communication
- Real-time SOS emergency alerts
- Opportunistic cloud synchronization
- Centralized monitoring dashboard
- Improved traveller safety